Test Project

Cloud Computing

JP-KR Samsung Skills Friendship Match

Day 1

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Contents

This Test Project consists of the following documentation/files:

1. SFC2026_TP53_day1_actual_en
2. SFC2026_TP53_day2_actual_en
3. SFC2026_TP53_day3_actual_en
4. SFC2026_TP53_day4_actual_en
5. Binaries in the s3 bucket

Description of project

Goals

This Test Project aims to rigorously evaluate your cloud computing skills through a series of challenges.

Background

You've been hired as an engineer to design architecture and operate infrastructure. You're tasked with managing computing resources operating in both the company's IDC environment and the cloud vendor AWS environment. You're responsible for enabling connectivity between these computing resources. You're also responsible for managing the company's compliance and ensuring that internal computing resources undergo compliance audits. Refer to the requirements and perform detailed tasks. Your CEO is concerned that operating costs will skyrocket as the network size increases. Be careful not to create unnecessary resources beyond the requirements.

Caution

1. Be careful not to create unnecessary resources beyond the requirements.
2. All resources, except global resources, use the Virginia (us-east-1) region unless otherwise specified.
3. All EC2 instances use the t3.micro instance type unless otherwise specified.
4. All EC2 instances must enable TCP 80/443 outbound through anyopen.
5. All EC2 instances must be accessible through SSM Session Manager.
6. Values not provided in the test project should be set based on the AWS Well-Architected Framework 6 pillars.
7. Creating unnecessary resources, even if they do not incur costs, may result in a score penalty.
   (e.g., creating additional VPCs)

8) Never use Marketplace AMIs.

Initial State

The AWS account provided does not come with any initial configuration. Refer to the requirements to create and modify the necessary resources and configure the assignment.

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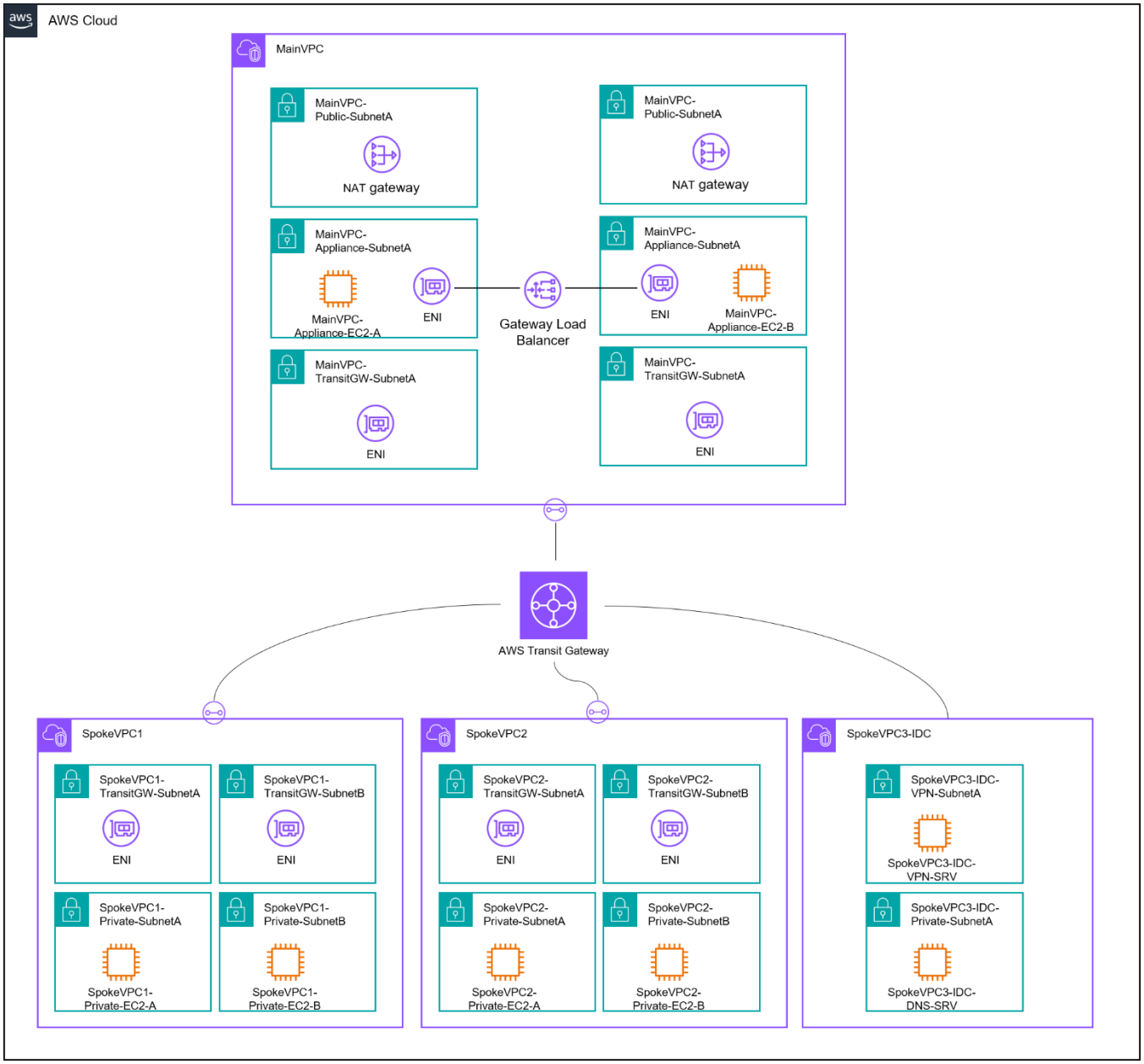
Architecture

Please refer to the architecture below for your task. The architecture is for reference only; you should configure the details to requirements.



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Technical Details

Step 1: Implement base networks on VPC

Perform VPC configuration tasks as follows. Some tasks that are not specified should be performed based on other requirements. *MainVPC-Public-SubnetA* and *MainVPC-Public-SubnetB* must be accessible to external networks through the Internet Gateway.

VPC Information

Name Tag	CIDR
MainVPC	10.0.0.0/16
SpokeVPC1	10.1.0.0/16
SpokeVPC2	10.2.0.0/16
SpokeVPC3-IDC	10.3.0.0/16

Subnets Information

VPC	Name Tag	CIDR	Availability Zone
MainVPC	MainVPC-Public-SubnetA	10.0.1.0/24	us-east-1a
	MainVPC-Public-SubnetB	10.0.2.0/24	us-east-1b
	MainVPC-Appliance-SubnetA	10.0.3.0/24	us-east-1a
	MainVPC-Appliance-SubnetB	10.0.4.0/24	us-east-1b
	MainVPC-TransitGW-SubnetA	10.0.5.0/24	us-east-1a
	MainVPC-TransitGW-SubnetB	10.0.6.0/24	us-east-1b
SpokeVPC1	SpokeVPC1-Private-SubnetA	10.1.1.0/24	us-east-1a
	SpokeVPC1-Private-SubnetB	10.1.2.0/24	us-east-1b
	SpokeVPC1-TransitGW-SubnetA	10.1.3.0/24	us-east-1a
	SpokeVPC1-TransitGW-SubnetB	10.1.4.0/24	us-east-1b
SpokeVPC2	SpokeVPC2-Private-SubnetA	10.2.1.0/24	us-east-1a
	SpokeVPC2-Private-SubnetB	10.2.2.0/24	us-east-1b
	SpokeVPC2-TransitGW-SubnetA	10.2.3.0/24	us-east-1a
	SpokeVPC2-TransitGW-SubnetB	10.2.4.0/24	us-east-1b
SpokeVPC3-IDC	SpokeVPC3-IDC-Private-SubnetA	10.3.1.0/24	us-east-1a
	SpokeVPC3-IDC-VPN-SubnetA	10.3.2.0/24	us-east-1b

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Step 2: Launch EC2 instances

Configure your EC2 instances using the table below. All EC2 instances must allow all ICMP protocols in their inbound and outbound rules. All EC2 instances must have network tools installed for testing and marking, allowing the use of the *ping*, *tracert*, and *tcpdump* commands.

Name Tag	Located Subnet
MainVPC-Appliance-EC2-A	MainVPC-Appliance-SubnetA
MainVPC-Appliance-EC2-B	MainVPC-Appliance-SubnetB
SpokeVPC1-Private-EC2-A	SpokeVPC1-Private-SubnetA
SpokeVPC1-Private-EC2-B	SpokeVPC1-Private-SubnetB
SpokeVPC2-Private-EC2-A	SpokeVPC2-Private-SubnetA
SpokeVPC2-Private-EC2-B	SpokeVPC2-Private-SubnetB
SpokeVPC3-IDC-VPN-SRV	SpokeVPC3-IDC-VPN-SubnetA
SpokeVPC3-IDC-DNS-SRV	SpokeVPC3-IDC-Private-SubnetA

Step 3: Make connectivity between each VPC using Transit Gateway

To ensure compliance, configure the network in a hub-and-spoke configuration using a Transit Gateway. Create a single Transit Gateway and create attachments for *MainVPC*, *SpokeVPC1*, and *SpokeVPC2*. Configure the ENIs for each VPC attachment to be located in the *TransitGW-Subnet*. *SpokeVPC3-IDC* should not be directly attached to the Transit Gateway; instead, it should be connected to the TransitGateway network via a Site-to-Site VPN. For detailed configuration, refer to Site-to-Site VPN.

SpokeVPC1, *SpokeVPC2*, and *SpokeVPC3-IDC* must be able to communicate with each other and externally through *MainVPC*. External communication must be enabled through a NAT Gateway located in Public-Subnet of *MainVPC*. *SpokeVPC1* and *SpokeVPC2* do not need to communicate with each other. Configure routing to prevent communication between them.

Step 4: Connect network between Cloud environment and IDC environment using Site-to-Site VPN

To connect the SpokeVPC3-IDC network and the TransitGateway network, configure a Site-to-Site VPN. This task is performed using the SpokeVPC3-IDC-VPN-SRV instance located in SpokeVPC3-IDC-VPN-SubnetA as the VPN server. We recommend using a tool such as StrongSwan or Quagga to configure the VPN. However, you can also use other tools to configure the VPN.

Step 5: Query to domain name of IDC

SpokeVPC3-IDC-DNS-SRV is configured to provide DNS services via BIND9. The DNS server must provide following record information without restrictions. Then, *SpokeVPC1* and *SpokeVPC2* to use the DNS services provided by *SpokeVPC3-IDC*. Existing DNS queries should also work without issues. For example, queries to *google.com* should also be possible.

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Domain Name	Record Type	Address
www.skill53.local	A	10.3.2.220
malicious.skill53.local	A	10.3.2.224

Step 6: Ensure the compliance using Gateway Load Balancer

Configure *MainVPC-Appliance-EC2-A* and *MainVPC-Appliance-EC2-B* to ensure compliance, including capturing network packets for incoming and outgoing traffic from the internal network. Do not use Marketplace AMIs. We recommend using *Suricata*, an open-source network security tool. Other third-party tools are also acceptable. Configure a Gateway Load Balancer to ensure proper compliance. The compliance tool is not required to perform any actions; it has to simply captures traffic.

Step 7: Query exception handling on IDC

SpokeVPC3-IDC-DNS-SRV should provide DNS services via BIND9 and store query logs in */var/log/named/queries.log* . *malicious.skill53.local* is a malicious domain name record that was not intended by the engineer. If a query request for malicious domain name record is written to the query log, the log should be stored in */var/log/named/malicious.log* in the following format. **Note! Since this server is assumed to be an IDC server, AWS products such as EventBridge and Lambda cannot be used to troubleshoot this issue.**

```
'{"query": "<Query Domain Name>", "Record": "<Record Type>", "Address": "<Address>", "Client": "<Client IP Address>"}'
```


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